

Experiential Learning Internship for Teaching Design Research

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Abstract

HCI researchers have explored ways to improve HCI and design education. We designed and conducted an experiential learning internship to teach principles of design research to first-year undergraduate students. We describe the framework we used for the design of the internship and the lessons we learned in a self-reflective case study. We provide the following recommendations to other educators who want to incorporate similar programs: 1) prepare for clients having differences in opinion, 2) use multiple educational methods to supplement the experience, 3) allow interns to navigate a project space on their own, but identify areas that may create major delays, and 4) prepare educational material on the latest trends and tools that may be of interest to the interns.

CCS Concepts

• **Applied computing** → *Interactive learning environments*; • **Human-centered computing** → *User centered design*; • **Social and professional topics** → *Model curricula*.

Keywords

Design research, Experiential learning, HCI Education

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1 Introduction

This case study examines an innovation-focused internship that uses design research to teach first-year undergraduate students practical and professional skills. Through the internship, students gain experience using design techniques, including primary and

secondary research to develop and test hypotheses, learn how to bridge the gap between qualitative and quantitative research, build a practice of community engagement, and focus on the humanistic foundations of research while critically utilizing the latest technological innovations. Local organizations, businesses, and startups, referred to as community partners in this paper, identified projects for the interns to work on. This case study presents the internship framework, lessons learned, and applications for innovation and design research programs.

1.1 Design Research Education

HCI and interaction design have both theoretical and practical aspects and range multiple disciplines, which can make them difficult to grasp. Prior HCI scholars have described efforts to enhance the efficacy of design education [13, 16, 20, 38, 43, 45, 48]. However, more investigation is needed for our understanding of design research education in and out of the classroom.

The internship which serves as the setting for our observations, was grounded in its development in the concerns of student affairs and how to support student development. Developed as a program within the Innovation, Design, and Entrepreneurship Academy (IDEA), a special initiative of the Office of the Provost, it was interested in developing a model for interdisciplinary experiential learning opportunities with a centrally located mediator that could guide first-year students on how to leverage university resources during their college journey. This new program used innovation, encompassing design and entrepreneurial practices, as its thematic focus.

Innovation and entrepreneurship programs grew rapidly at universities in the early 21st century as organizations increasingly turned to the notions of innovation and design thinking to solve complex societal issues [5, 15, 23, 26]. University innovation and entrepreneurship programs have several key elements: focus on undergraduate students, experience of flexible space, a systemic approach to problem solving (design thinking), and university-industry partnerships [33, 36, 40]. Innovation and entrepreneurship programs are also touted for their value for workforce development, in particular 21st century skills [14, 33].

IDEA was initially informed by Stanford d.School's practice of Design Thinking drawing from experiential and problem-based learning [7, 18, 22, 28, 47], design pedagogy from the field of Engineering [2, 6, 11, 19, 35, 39], and cognitive constructivism [3, 41].

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Brown, [4] through his work with Stanford's d.School and the company IDEO, has promoted a methodology of human-centered design formed from the Engineering tradition of design that cuts across contexts [26, 27]. This conceptualization of human-centered design is situated among a plurality of discourses that seek to define it: in individual disciplines such as Engineering, Fine Arts, Architecture, Computer Science, and Information Science; in the liberal arts tradition where design thinking integrates the technological, human experience, and artifact [5]; a general theory of design [37]; an organizational resource for innovation [26]; and in the neoliberal notion that an individual's values, when applied for the purpose of taking action for the benefit of others, can lead to the best outcome for society¹.

1.2 Workforce Development

Workforce development, specifically 21st Century skills training, is of particular focus at state and federal agencies [1, 12, 17, 24, 25, 31]. 21st Century Skills broadly include soft skills like critical thinking, problem-solving, collaboration, and creativity that can be found across disciplines; as well as hard skills that are discipline-specific [14, 32, 40]. Universities have long used experiential learning in the form of internships and coops to provide students with realistic opportunities to develop and practice 21st Century skills. Design research internships are one such opportunity where students can practice soft and hard skills.

1.3 Experiential Learning

In the liberal arts tradition, design thinking can be used to operationalize constructivist learning theory, seen in several key elements: involvement of students, experience of flexible space, systemic approach to problem solving, and a balance between open and closed (detailed) instruction [40]. The theoretical perspectives that inform design thinking in the liberal arts tradition are summarized by Johansson-Sköldberg et al (2013) [21] as "(1) *Design and designerly thinking as creation of artifacts* [44]; (2) *Design and designerly thinking as a reflexive practice* [42]; (3) *Design and designerly thinking as a problem-solving activity* [5]; (4) *Design and designerly thinking as a way of reasoning/making sense of things* [8, 9, 30]; (5) *Design and designerly thinking as creation of meaning* [29]." (p.124). These theoretical perspectives overlap with the theoretical perspectives of experiential learning: (1) Learning as a continuous, social process where new experiences are integrated with older ones [10, 46]; and (2) Knowledge created through a learning process that incorporates experiencing, reflecting, thinking, and iterating [28].

While research has shown the benefits of incorporating experiential learning into classroom activities, few describe the practical application of using experience in teaching design research. For instance, Yang et al. [48] describe a theoretical approach to using experiential learning in a classroom setting, but could be furthered through investigating experiential learning in practice. Furthermore, while classroom settings can be beneficial for their controlled

environment, it is uncertain how the controlled classroom environment translates to dynamic environments of a workplace setting. This paper seeks to describe the author's reflections on their experience in conducting a summer internship program for teaching design research to undergraduate students, combining traditional class experience with guided practical work experience.

2 Internship Framework

2.1 Structure

The internship was structured to be the first professional experience for rising Sophomores, coming at the end of their first year at university. It was based on the belief that early engagement in real-world work settings allow first- and second-year students to explore their interests, learn workplace etiquette, and acquire key skills for their resumes. Whereas industry-sponsored internships are typically reserved for rising Juniors and Seniors as it allows employers to assess prospective employees and upperclassmen have more experience by virtue of having completed more upper level courses and have identified a chosen career path; students want to start earlier with connecting their classroom learning with practical work experience.

In order to provide first-year students an opportunity for an early professional experience the internship was structured as a consulting and design agency where IDEA solicited projects from community organizations and recruited interdisciplinary teams of student interns to work on the projects. The types of projects IDEA accepted for the internship program articulated a clear purpose with a recognized value to the organization or society; had specific functional objectives with a scope broad enough to provide opportunity for exploration through design methods.

IDEA partnered with Hajar Shirley, Research Director at Inclusive Health Innovation Together, to create an agile framework to organize and structure the experience. The ten-week program was divided into short, time-boxed periods where the design process was modeled and interns were guided on how to use specific methods and apply them with their project. As part of the program, interns were expected to conduct 1-2 site visits each with their community partner: "*Borrowing from Anthropology, a researcher's field site is their source of observing and collecting data. Fieldwork is the process of immersing oneself in as many aspects of the daily cultural lives of people as possible in order to study their behaviors and interactions. Nearly any setting or location can become "the field": a village along the Amazon River, a large corporate office, a small neighborhood café, or even a social networking site like Facebook.*" [34].

2.2 Research Methods Workshops

Weekly workshops focused on exploring "*the ways research is imagined, understood, planned, and executed*" [34]. The purpose was to provide training on research methods so that students could build skills and a methods toolkit (see Table 01.)

3 Lessons Learned

To provide a better educational experience for the interns, the authors conducted a self-analysis of the program to identify areas for continuous improvement. Rutgers University Institutional Review

¹Dr. Carolyn Moehling expressed this idea of applying values for societal good and the need for innovative thinking during a personal conversation with the second author on March 6, 2024. Dr. Moehling is the Senior Vice Provost and Vice Chancellor for Undergraduate Education at Rutgers University-New Brunswick

Table 1: IDEA Design Research Internship Framework

Week	Theme	Sample Topics
1	Onboarding Engage	Community engagement: People, systems, & roles
2	Explore & Connect (Empathy)	Qualitative data, Data collection, Conducting fieldwork
3	Explore & Connect (Empathy)	What does it mean to be participatory in design research?
4	Analyze & Reimagine (Define, Ideate)	Research and storytelling: Narratives, coding, analysis
5	Analyze & Reimagine (Define, Ideate)	Reframing a project
6	Making & Testing (Prototype, Test)	Types of prototypes and when to use them
7	Making & Testing (Prototype, Test)	Pilot studies, usability testing, and evaluation
8	Reflection	Next steps and project handoff
9	Reporting Findings & Exiting (Storytelling)	Presenting findings in Design Research
10	Reporting Findings & Exiting (Storytelling)	Presentations

Board reviewed this study’s protocol numbered Pro2024002175. In this section, we highlight our lessons learned in conducting the internship program, the challenges and facilitators of internship education, and our recommendations for how others can incorporate similar experiences into their own programs.

3.1 Teaching Design Process

Since most of our interns did not come from a specific interaction design program, but rather came from a variety of disciplines such as computer science, graphic design, and business, we provided workshops as a way to educate the interns on the design process. Although the hope was that the interns would gain practical experience through their projects, the workshops provided a way for the interns to think about their projects in a formalized design process. This was beneficial in cases where clients would rather take different approaches to the projects. For instance, two of the clients took an approach of wanting to build their product and find the issues later. Through the workshops we were able to show where user research would take place and what it would incorporate while still providing the clients the deliverables they were looking for.

In a similar case, one of the intern teams immediately began using Figma to prototype their product as one of the first steps of their project. During our mentorship sessions, we reminded them of the initial steps of the process and recommended they go backwards and see what new information they found. In doing so, the interns discovered in user research that the users "did not want another program" but instead wanted a more streamlined system. In this example, the client wanted a program to provide recommendations to help users find courses of interest. However, the course selection process already involved three programs so adding another for a different purpose would only add to the cognitive load of users. Because the interns were able to come up with their own plan of action, make mistakes in a low stake environment, and had guidance from the program mentors, the experience created a stronger lesson for the interns than simply telling them the importance of user research. The intern group then was able to share this experience with the other groups during our presentation sessions. Since each project had different experiences, sharing the experiences and their own lessons learned in presentations gave the interns a more holistic view.

3.2 Navigating Ambiguity

As part of the program, we wanted to allow the interns to be able to gain experience as autonomous colleagues who are able to engage with their projects as they would in a workplace setting. However, as undergraduates, they are used to being told what to do and are most comfortable with clearly defined expectations. In one of the groups, the client was often unclear and non-specific as to what they wanted from the interns. As a result, the interns of that client thought they had nothing to do. Seeing this as an opportunity to learn, we guided the interns to conduct a brainstorming activity of what they could do. We then facilitated conversations with the client for the interns to give recommendations for their next steps rather than waiting to be told what to do.

We also wanted to give the interns an opportunity to experience the dynamic nature of a work environment. Unlike a class assignment, a client project often does not come with all the necessities well-defined. Interns needed to learn how to navigate project ambiguity by having discussions with the client and conducting user research to elicit requirements. While interns were often effective at navigating this space, there were also issues when the client either did not communicate a change in the project scope in the time between their project proposal and the internship or did not specify specific equipment that was needed. One of the project groups was significantly delayed because the client had not specified that the prototype was expected to run fully on a Virtual Reality Headset. While the program and hardware needed were easily accessible, being in an academic system required approvals and re-imaging of the computers in order to have the programs to create the prototype. Since the program had a defined end, any extended delay was significant. This highlighted to us the importance of probing the clients further in the time before the program start. While we still want to give the interns the experience of navigating ambiguity, we recognize the need to avoid extended delays due to administrative requirements. This project also allowed for us to give the interns a clear example of why low-fidelity prototyping is valuable, since we had a clear example of how long a high fidelity prototype can take.

3.3 Intern Interests

Throughout the program, the interns were highly engaged with the topics being discussed. It is well-recognized that students are better engaged with concepts that relate back to real-life situations.

In this way, the project themselves generated interest as they were real projects from businesses and community partner organizations. However, interns also wanted to know about the latest advances in technology and their application to design research. In this program, interns often asked about how AI could be used since generative AI was generating mass discourse in the media and on campus regarding academic integrity. They were interested in exploring its potential while also cognizant of its issues. It was important for us to be prepared to discuss how AI is being used in design spaces, the opportunities they brought, and the risks associated with AI. We initially did not expect to discuss AI in any great detail, but when the interns showed interest in the topic, we prepared workshops and activities that covered opportunities (e.g. brainstorming and modeling activities) and considerations (e.g. privacy concerns, outdated information in generative models, and poor quality information).

Interns were also highly engaged in an end-of-program poster session that was held jointly with other summer research programs. This gave the interns an opportunity to reflect on how much they had accomplished in a short time, the lessons that they had learned, and was a source of pride in the work they had done. This poster session allowed for a final creative outlet for the interns and incorporated a sense of completion which was beneficial for those who needed to hand off in progress work to the community partners.

4 Conclusion

Experiential learning has benefits in its ability to generate a construction of knowledge in students. We designed a summer internship that leveraged experience through the cooperation of project sponsors, combined with guided mentorship and workshops to provide a comprehensive learning experience of design research. However, the nature of conducting projects in the real world and working with businesses and community partners outside academia comes with challenges for ensuring an environment conducive to learning. We describe lessons that we learned through a self-reflection of the program and provide the following recommendations to other educators who want to incorporate similar programs: 1) prepare for clients having differences in opinion, 2) use multiple educational methods to supplement the experience such as workshops, presentations, and poster sessions, 3) allow interns to navigate a project space on their own, but identify areas that may create major delays, 4) prepare educational material on the latest trends and tools that may be of interest to the interns. Despite the challenges that HCI and design educators may face, an experiential learning program better prepares interns to apply the knowledge acquired during their undergraduate education in a professional setting prior to entering the workforce.

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References

- [1] Kathryn Anderson-Levitt. 2021. 21st century skills in the United States: a late, partial and silent reform. *Comparative Education* 57, 1 (Jan. 2021), 99–114. <https://doi.org/10.1080/03050068.2020.1845059>
- [2] Sara L. Beckman and Michael Barry. 2007. Innovation as a Learning Process: Embedding Design Thinking. *California Management Review* 50, 1 (Oct. 2007), 25–56. <https://doi.org/10.2307/41166415> Publisher: SAGE Publications Inc.
- [3] Elizabeth Bonawitz, Patrick Shafto, Hyowon Gweon, Noah D. Goodman, Elizabeth Spelke, and Laura Schulz. 2011. The double-edged sword of pedagogy: Instruction limits spontaneous exploration and discovery. *Cognition* 120, 3 (Sept. 2011), 322–330. <https://doi.org/10.1016/j.cognition.2010.10.001>
- [4] Tim Brown. 2009. *Change by design: how design thinking transforms organizations and inspires innovation* (1. ed ed.). Harper Business, New York, NY.
- [5] Richard Buchanan. 1992. Wicked Problems in Design Thinking. *Design Issues* 8, 2 (1992), 5–21. <https://doi.org/10.2307/1511637> Publisher: The MIT Press.
- [6] L. Cavagnaro and H. Fasihuddin. 2016. A moonshot approach to change in higher education. *Liberal Education* 102 (2016), 8–17.
- [7] Alison Cook-Sather. 2012. Lessons in Higher Education: Five Pedagogical Practices that Promote Active Learning for Faculty and Students. *Journal of Faculty Development* 26, 1 (2012), 1–7.
- [8] N. Cross. 2006. *Designing Ways of Knowing*. Springer Verlag, London.
- [9] N. Cross. 2011. *Design Thinking*. Berg, Oxford.
- [10] John Dewey. 1923. Social purposes in education. *General Science Quarterly* 7, 2 (Jan. 1923), 79–91. <https://doi.org/10.1002/scs.3730070201>
- [11] Clive L. Dym, Alice M. Agogino, Ozgur Eris, Daniel D. Frey, and Larry J. Leifer. 2005. Engineering Design Thinking, Teaching, and Learning. *Journal of Engineering Education* 94, 1 (2005), 103–120. <https://doi.org/10.1002/j.2168-9830.2005.tb00832.x> eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1002/j.2168-9830.2005.tb00832.x>
- [12] Adam K. Edgerton. 2022. *Strengthening Career and Technical Education for the 21st Century Act (Perkins V): A Primer*. Technical Report CRS Report R47071. Congressional Research Service. <https://crsreports.congress.gov/product/pdf/R/R47071>
- [13] Hai-Guang Fang, Ying-Wen-Hui Huang, Wen-Qi Man, and Xin-Mei Kong. 2023. Research on teaching design framework and evaluation using graphical programming towards cultivating students' digital competence. In *2023 7th International Conference on Digital Technology in Education*. ACM, Hangzhou China, 129–133. <https://doi.org/10.1145/3626686.3631652>
- [14] Abdul Ghafar. 2020. Convergence between 21st Century Skills and Entrepreneurship Education in Higher Education Institutes. *International Journal of Higher Education* 9, 1 (Jan. 2020), 218. <https://doi.org/10.5430/ijhe.v9n1p218>
- [15] Sharon Guaman-Quintanilla, Patricia Everaert, Katherine Chiliza, and Martin Valcke. 2023. Impact of design thinking in higher education: a multi-actor perspective on problem solving and creativity. *International Journal of Technology and Design Education* 33, 1 (March 2023), 217–240. <https://doi.org/10.1007/s10798-021-09724-z>
- [16] Alireza Hakamian and Steffen Becker. 2023. Quantitative Analysis of Software Designs: Teaching Design and Experiences. In *Companion of the 2023 ACM/SPEC International Conference on Performance Engineering*. ACM, Coimbra Portugal, 341–344. <https://doi.org/10.1145/3578245.3584357>
- [17] Sara Haviland and Steven Robbins. 2021. Career and Technical Education as a Conduit for Skilled Technical Careers: A Targeted Research Review and Framework for Future Research. *ETS Research Report Series* 2021, 1 (Dec. 2021), 1–42. <https://doi.org/10.1002/ets2.12318>
- [18] Jonathan Hey, Alan Van Pelt, Alice Agogino, and Sara Beckman. 2007. Self-Reflection: Lessons Learned in a New Product Development Class. *Journal of Mechanical Design* 129, 7 (July 2007), 668–676. <https://doi.org/10.1115/1.2722781>
- [19] Jonathan H. G. Hey and Alice M. Agogino. 2007. Metaphors in Conceptual Design. In *Volume 3: 19th International Conference on Design Theory and Methodology; 1st International Conference on Micro- and Nanosystems; and 9th International Conference on Advanced Vehicle Tire Technologies, Parts A and B*. ASMEDC, Las Vegas, Nevada, USA, 125–134. <https://doi.org/10.1115/DETC2007-34874>
- [20] Ping Jian, De-sheng Liu, and Fan-de Yang. 2024. Research on Teaching Design of "System of Systems Design and Evaluation" Course Based on System Thinking. In *Proceedings of the 2024 Guangdong-Hong Kong-Macao Greater Bay Area International Conference on Education Digitalization and Computer Science*. ACM, Shenzhen China, 50–55. <https://doi.org/10.1145/3686424.3686433>

- [21] Ulla Johansson-Sköldberg, Jill Woodilla, and Mehves Çetinkaya. 2013. Design Thinking: Past, Present and Possible Futures. *Creativity and Innovation Management* 22, 2 (June 2013), 121–146. <https://doi.org/10.1111/caim.12023>
- [22] Manu Kapur and Katherine Bielaczyc. 2012. Designing for Productive Failure. *The Journal of the Learning Sciences* 21, 1 (2012), 45–83. <https://www.jstor.org/stable/23266307> Publisher: Taylor & Francis, Ltd..
- [23] Kauffman Foundation. 2013. *Entrepreneurship Comes of Age on Campus*. Technical Report. Kauffman, The Foundation of Entrepreneurship. https://www.kauffman.org/wp-content/uploads/2019/12/eshipedcomesofage_report.pdf
- [24] Robert Kelchen, Justin Ortagus, Kelly Rosinger, Dominique Baker, and Mitch Lingo. 2024. The Relationships Between State Higher Education Funding Strategies and College Access and Success. *Educational Researcher* 53, 2 (March 2024), 100–110. <https://doi.org/10.3102/0013189X231208964>
- [25] Robert Kelchen, Justin Ortagus, Kelly Rosinger, and Alex Cassell. 2024. Investing in the Workforce: The Impact of Performance-Based Funding on Student Earnings Outcomes. *The Journal of Higher Education* 95, 2 (Feb. 2024), 172–199. <https://doi.org/10.1080/00221546.2023.2171201>
- [26] Lucy Kimbell. 2011. Rethinking Design Thinking: Part I. *Design and Culture* 3, 3 (Nov. 2011), 285–306. <https://doi.org/10.2752/175470811X13071166525216>
- [27] Lucy Kimbell. 2012. Rethinking Design Thinking: Part II. *Design and Culture* 4, 2 (July 2012), 129–148. <https://doi.org/10.2752/175470812X13281948975413>
- [28] Alice Y. Kolb and David A. Kolb. 2005. Learning Styles and Learning Spaces: Enhancing Experiential Learning in Higher Education. *Academy of Management Learning & Education* 4, 2 (June 2005), 193–212. <https://doi.org/10.5465/amle.2005.17268566>
- [29] K Krippendorff. 2006. *The Semantic Turn: A New Foundation for Design*. Taylor & Francis, Boca Raton, FL.
- [30] Bryan Lawson. 2010. *How designers think: the design process demystified* (reprint ed.). Elsevier Architectural Press, Amsterdam Heidelberg.
- [31] Lewis-Burke Associates LLC. 2021. Lewis-Burke Analysis of the Federal Innovation and Economic Development Landscape.
- [32] J.K. McDonald, R.E. West, P.J. Rich, and I. Pfleger. 2019. “It’s So Wonderful Having Different Majors Working Together”: the Development of an Interdisciplinary Design Thinking Minor. *TechTrends* 63, 4 (2019), 440–450. <https://doi.org/10.1007/s11528-018-0325-2>
- [33] Ghulam Nabi, Francisco Liñán, Alain Fayolle, Norris Krueger, and Andreas Walmesley. 2017. The Impact of Entrepreneurship Education in Higher Education: A Systematic Review and Research Agenda. *Academy of Management Learning & Education* 16, 2 (June 2017), 277–299. <https://doi.org/10.5465/amle.2015.0026>
- [34] Zainab Najeeb and Veronica Armour. 2024. 2024 Internship Program Map.
- [35] Dev Patnaik and Robert Becker. 1999. Needfinding: The Why and How of Uncovering People’s Needs. *Design Management Journal (Former Series)* 10, 2 (1999), 37–43. <https://doi.org/10.1111/j.1948-7169.1999.tb00250.x> <https://onlinelibrary.wiley.com/doi/pdf/10.1111/j.1948-7169.1999.tb00250.x>
- [36] Walter W. Powell, Kenneth W. Koput, and Laurel Smith-Doerr. 1996. Interorganizational Collaboration and the Locus of Innovation: Networks of Learning in Biotechnology. *Administrative Science Quarterly* 41, 1 (March 1996), 116. <https://doi.org/10.2307/2393988>
- [37] Johan Redström. 2017. *Making Design Theory*. MIT Press, Cambridge, MA, USA. <https://doi-org.proxy.libraries.rutgers.edu/10.7551/mitpress/11160.001.0001>
- [38] Emma J. Rose, Craig Macdonald, and Cynthia Putnam. 2023. Design Systems: A scalable model for teaching design systems for UX. In *Proceedings of the 5th Annual Symposium on HCI Education*. ACM, Hamburg Germany, 5–7. <https://doi.org/10.1145/3587399.3587403>
- [39] Adam Royalty, Lindsay Noelle Oishi, and Bernard Roth. 2014. Acting with Creative Confidence: Developing a Creative Agency Assessment Tool. In *Design Thinking Research: Building Innovation Eco-Systems*, Larry Leifer, Hasso Plattner, and Christoph Meinel (Eds.). Springer International Publishing, Cham, 79–96. https://doi.org/10.1007/978-3-319-01303-9_6
- [40] Andrea Scheer, Christine Noweski, and Christoph Meinel. 2012. Transforming Constructivist Learning into Action: Design Thinking in Education. *Design and Technology Education* 17, 3 (2012), 8–19. <https://eric.ed.gov/?id=EJ996067> Publisher: Design and Technology Education Association.
- [41] Alan H. Schoenfeld. 2016. What’s All the Fuss about Metacognition? In *Cognitive science and mathematics education*, Alan H. Schoenfeld (Ed.). RoutledgeTaylor & Francis Group, London, 189–215.
- [42] Donald A. Schön. 1983. *The reflective practitioner: how professionals think in action*. Basic Books, New York.
- [43] Dan Shi. 2022. Teaching Design Based on Online Collaborative Learning. In *Proceedings of the 7th International Conference on Distance Education and Learning*. ACM, Beijing China, 122–126. <https://doi.org/10.1145/3543321.3543341>
- [44] Herbert Alexander Simon. 1969. *The sciences of the artificial* (3. ed., [nachdr.] ed.). MIT Press, Cambridge, Mass. OCLC: 552080160.
- [45] Hoa Vo. 2024. Teaching Design with Consumer-ready Artificial Intelligence and Virtual Reality: A Case Study. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Educator’s Forum*. ACM, Denver CO USA, 1–2. <https://doi.org/10.1145/3641235.3664437>
- [46] Lev Semenovič Vygotskij, Michael Cole, and Vera John-Steiner. 1978. *Mind in society: the development of higher psychological processes* (nachdr. ed.). Harvard Univ. Press, Cambridge, Mass.
- [47] Aman Yadav, Dipendra Subedi, Mary A. Lundeborg, and Charles F. Bunting. 2011. Problem-based Learning: Influence on Students’ Learning in an Electrical Engineering Course. *Journal of Engineering Education* 100, 2 (2011), 253–280. <https://doi.org/10.1002/j.2168-9830.2011.tb00013.x> <https://onlinelibrary.wiley.com/doi/pdf/10.1002/j.2168-9830.2011.tb00013.x>
- [48] Lihao Yang, Qi Zhang, and Huafeng He. 2023. Research on Teaching Design Method Based on L-Kolb Cycle. In *Proceedings of the 2023 International Conference on Information Education and Artificial Intelligence*. ACM, Xiamen China, 105–109. <https://doi.org/10.1145/3660043.3660062>